

HcZZ ctag2d -- Results (official SFs, L0 swap reverted 2026-09-15)

Production: combine_run3_100150_ctag2d_v8_jecshifts_ggzzkfactor_nox_rawsf
hplusc_mva_4class_ctag2d_scored_v8 + hplusc_mva_4class_ctag2d_jecshifts_scored_v2,
[100,150] GeV, no-ZX (MC-only), 15 bins (quantile + surgical rebin "2-4,7-10"),
23 nuisances. CMS_ctag2d uses the OFFICIAL up_Total/down_Total, unmodified --
the 2026-09-02 swap correction was retracted 2026-09-15 (component-level
re-investigation found its basis too thin; reported to the calibration's
maintainer, not yet POG-confirmed either way). Rebuilt and verified 2026-09-15.

Yields [100,150] GeV

Process	Rate
Signal	0.053130
ggZZ	6.677900
qqZZ	36.318223
Other_Higgs	55.077609

$S/\sqrt{B} = 0.0054$

Final limits (AsymptoticLimits, --run blind)

Quantile	r	kappa_c
2.5%	164.2617	31.13
16%	223.0386	41.34
50% (median)	321.0000	58.33
84%	477.1014	85.39
97.5%	687.6224	121.87

Stat-only (--freezeParameters allConstrainedNuisances): r=273.5/kappa_c=50.09 --
identical to the retracted-fix version (only CMS_ctag2d's shape changed, not
central yields). Without the L0 swap correction: r=321.0 (was 411.0 with it),
a -21.9% shift -- the single largest correction to this headline's history.

CMS_ctag2d dominates the fit (~4.1x the #2 nuisance, down from ~8.6x under the
retracted swap correction). JES/JER's 4 rows (CMS_scale_j/CMS_res_j/CMS_scale_m/
CMS_res_m) rank #3/#15/#17/#16 by impact -- CMS_scale_j in particular is now
the #3 nuisance overall, more prominent than under the retracted fix (#5).

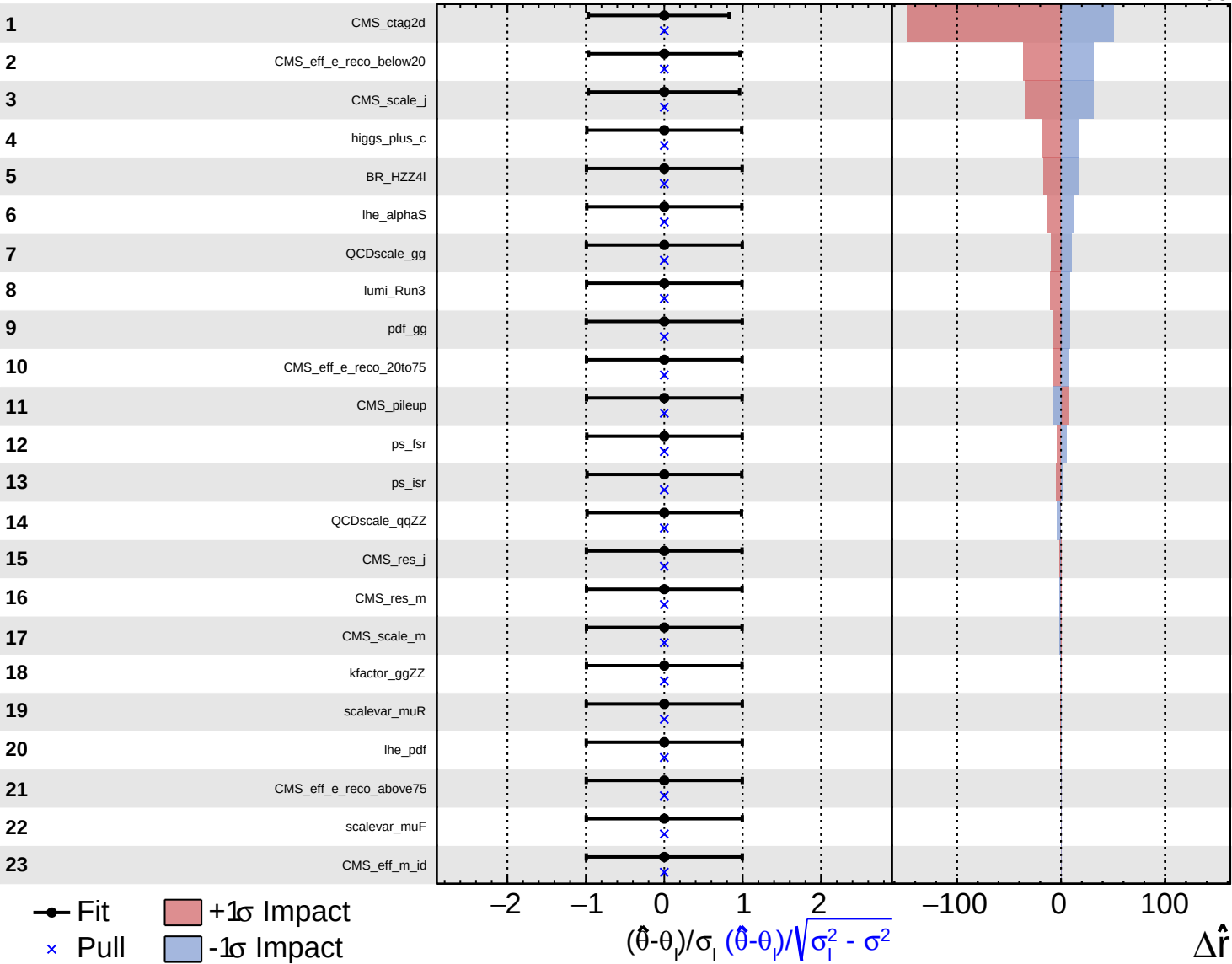
Systematics: type, assigned value, and impact

Ranked by impact_r (combineTool.py -M Impacts, --expect-signal 321.0, official ctag2d SFs, 23 nuisances). InN rows show the assigned size per affected process; shape rows show the post-fit pull instead (all prefit +/-1sigma here, expected for a blind Asimov fit).

#	Systematic	Type	Value (InN size / pull)	Impact
1	CMS_ctag2d	shape	pull -0.00 (-0.969/+0.826)	147.74
2	CMS_eff_e_reco_below20	shape	pull -0.00 (-0.967/+0.965)	36.39
3	CMS_scale_j	shape	pull +0.00 (-0.969/+0.960)	34.27
4	higgs_plus_c	shape	pull +0.00 (-0.987/+0.987)	17.66
5	BR_HZZ4l	InN	1.02 (Signal, Other_Higgs)	17.18
6	lhe_alphaS	shape	pull +0.00 (-0.989/+0.989)	12.60
7	QCDscale_gg	InN	1.039 (Signal, ggZZ)	10.38
8	lumi_Run3	InN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs)	10.26
9	pdf_gg	InN	1.032 (Signal, ggZZ)	8.48
10	CMS_eff_e_reco_20to75	shape	pull +0.00 (-0.992/+0.992)	7.75
11	CMS_pileup	shape	pull -0.00 (-0.992/+0.992)	7.43
12	ps_fsr	shape	pull +0.00 (-0.988/+0.993)	5.59
13	ps_isr	shape	pull -0.00 (-0.984/+0.988)	4.37
14	QCDscale_qqZZ	InN	1.04 (qqZZ)	3.45
15	CMS_res_j	shape	pull +0.00 (-0.993/+0.991)	1.67
16	CMS_res_m	shape	pull +0.00 (-0.992/+0.992)	1.12
17	CMS_scale_m	shape	pull -0.00 (-0.992/+0.992)	1.06
18	kfactor_ggZZ	InN	1.10 (ggZZ)	0.68
19	scalevar_muR	shape	pull +0.00 (-0.994/+0.994)	0.39
20	lhe_pdf	shape	pull -0.00 (-0.994/+0.994)	0.35
21	CMS_eff_e_reco_above75	shape	pull -0.00 (-0.994/+0.994)	0.12
22	scalevar_muF	shape	pull -0.00 (-0.994/+0.994)	0.08
23	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.03

CMS Internal

$\hat{r} = 320^{+200}_{-200}$



HcZZ -- with Z+X (true 4-era, pkatris) results, official SFs

Production: combine_run3_100150_ctag2d_v8_jecshifts_with_zx_pkatris_ggzzkfactor_rawsf [100,150] GeV, ctag2d scheme, 25 nuisances (18 shape/lnN + 4 JES/JER + ZX_norm/ZX_rate), true 4-era Z+X (pkatris production) via --zx-dir. Official ctag2d SFs, L0 swap correction retracted 2026-09-15 (see no-ZX page).

Yields in [100,150] GeV

Process	Yield
Signal	0.0531
ggZZ	6.6779
qqZZ	36.3182
Other_Higgs	55.0776
Z+X	133.2386
Total bkg	231.3123

$S/\sqrt{B} = 0.0035$

Final limits (combine -M AsymptoticLimits -m 120 --run blind)

Quantile	r	kappa_c
2.5%	167.0703	31.61
16%	227.7939	42.16
50% (median)	329.0000	59.72
84%	490.3032	87.68
97.5%	709.5030	125.66

Comparison against the no-ZX headline

Configuration	r (median)	kappa_c
no-Z+X headline (23 syst, official SFs)	321.0	58.33
with-Z+X, true 4-era (25 syst, official SFs)	329.0	59.72

Systematics with Z+X: type, assigned value, and impact

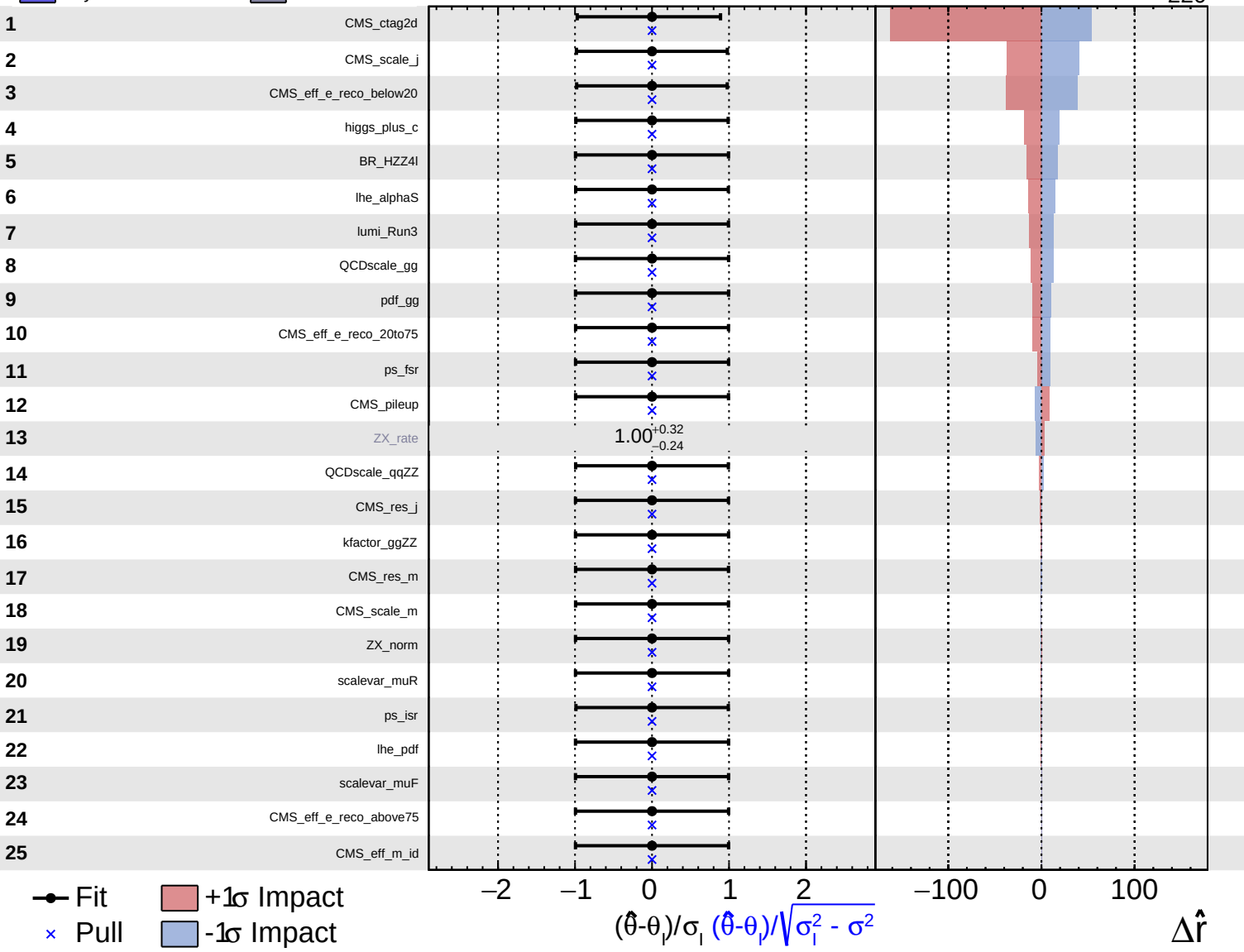
Ranked by impact_r (combineTool.py -M Impacts, --expect-signal 329.0, official ctag2d SFs, true 4-era Z+X included, 25 nuisances incl. the free-floating ZX_rate). ZX_rate is now #13 (was #2 under the retracted swap correction).

#	Systematic	Type	Value (lnN size / pull)	Impact
1	CMS_ctag2d	shape	pull -0.00 (-0.972/+0.890)	162.04
2	CMS_scale_j	shape	pull -0.00 (-0.982/+0.978)	40.41
3	CMS_eff_e_reco_below20	shape	pull -0.00 (-0.979/+0.977)	38.50
4	higgs_plus_c	shape	pull -0.00 (-0.990/+0.989)	18.76
5	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	17.68
6	lhe_alphaS	shape	pull -0.00 (-0.991/+0.991)	14.73
7	lumi_Run3	lnN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs, ZX)	12.83
8	QCDscale_gg	lnN	1.039 (Signal, ggZZ)	12.52
9	pdf_gg	lnN	1.032 (Signal, ggZZ)	10.20
10	CMS_eff_e_reco_20to75	shape	pull -0.00 (-0.993/+0.993)	9.57
11	ps_fsr	shape	pull +0.00 (-0.993/+0.994)	9.15
12	CMS_pileup	shape	pull +0.00 (-0.993/+0.993)	8.87
13	ZX_rate	rateParam	1.00 (+0.32/-0.24), free-floating	6.19
14	QCDscale_qqZZ	lnN	1.04 (qqZZ)	2.63
15	CMS_res_j	shape	pull -0.00 (-0.994/+0.993)	1.20
16	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.93
17	CMS_res_m	shape	pull +0.00 (-0.993/+0.993)	0.91
18	CMS_scale_m	shape	pull -0.00 (-0.993/+0.993)	0.83
19	ZX_norm	lnN	1.30 (ZX)	0.40
20	scalevar_muR	shape	pull +0.00 (-0.994/+0.994)	0.37
21	ps_isr	shape	pull +0.00 (-0.988/+0.991)	0.30
22	lhe_pdf	shape	pull -0.00 (-0.994/+0.994)	0.27
23	scalevar_muF	shape	pull -0.00 (-0.994/+0.994)	0.11
24	CMS_eff_e_reco_above75	shape	pull -0.00 (-0.994/+0.994)	0.05
25	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.02

Gaussian
 Poisson
 AsymmetricGaussian
 Unconstrained

CMS *Internal*

$\hat{r} = 330^{+200}_{-220}$



Stat-only limit: with vs. without Z+X (official SFs)

combine -M AsymptoticLimits --freezeParameters allConstrainedNuisances
(all constrained nuisances frozen; ZX_rate still floats since it is
unconstrained), same official-SF production, [100,150] GeV, both
re-verified 2026-09-15 -- IDENTICAL to the retracted-fix version, since the
swap correction only ever changed CMS_ctag2d's shape, never central yields.

Quantile	r (no ZX)	kc (no ZX)	r (with ZX)	kc (with ZX)	dr	dr %
2.5%	141.0234	27.08	145.0723	27.79	+4.05	+2.9%
16%	190.7021	35.72	198.0641	37.00	+7.36	+3.9%
50% (median)	273.5000	50.09	283.5000	51.83	+10.00	+3.7%
84%	396.6906	71.45	412.3249	74.16	+15.63	+3.9%
97.5%	553.4374	98.62	575.7133	102.48	+22.28	+4.0%

Key point: stat-only, Z+X loosens the median limit by only ~3.7%
(r=273.5 → 283.5) -- much smaller than the ~2.5% full-syst loosening
(r=321.0 → 329.0, official SFs). Both effects are now much smaller than
under the retracted swap correction (full-syst was +16.8%, r=411.0 → 480.0) --
with CMS_ctag2d no longer artificially inflated, Z+X's own systematics
(ZX_norm, ZX_rate) have much less room to loosen the fit further on top of
the stat-only effect.